Sahaj Bhadja

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Summary

Embedded software engineer with 2+ years building **C/C++ firmware**, **Linux/RTOS validation workflows**, and **hardware-software integration** across MCU protocols, automation pipelines, and safety-focused system testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **Python service orchestration** and robust data interfaces for 500+ users, reducing operational latency from **1.2s to 350ms** while improving observability for production issue triage.
- Implemented **fault-tolerant validation controls** and rollback safeguards for 5K+ monthly jobs, improving release reliability across distributed components and strict delivery timelines.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **React + Flask platform modules** handling 10K+ daily requests, converting requirements into reusable APIs/components with maintainable testing checkpoints for staged deployments.
- Improved **CI/CD reliability workflows** with automated verification and structured defect triage, reducing turnaround time by **60%** while preserving release quality under pressure.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **Python experimentation pipelines** for 100K+ trajectory logs, enabling reproducible benchmarking and disciplined result validation in safety-sensitive autonomous research workflows.
- Produced **debugging diagnostics** and technical summaries that translated findings into implementation plans, improving cross-team coordination between research and engineering groups.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **API/ETL microservices** using Python, AWS Glue, and S3, improving response times by **30%** while enabling dependable distributed data exchange for enterprise systems.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito across Linux/Windows, boosting pipeline efficiency by **68%** and reducing recurring production defects.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **MERN + WebSocket modules** supporting eight business workflows and 2M+ synchronized records, improving real-time consistency and resilience for customer-facing operations.
- Automated **Docker/Jenkins deployment pipelines**, reducing deployment times by **50%** and improving release repeatability through script-driven build and test gate controls.

Skills

Languages: **C/C++**, **Python**, JavaScript/TypeScript, Java, SQL, Bash
Embedded: **ARM Cortex MCUs**, RTOS, multithreading, GPIO, interrupts, timers, PWM
Protocols/Systems: **CAN**, **I2C**, **SPI**, UART, Linux/UNIX environments, device-level integration
Build & QA: Make/CMake, Jenkins, GitHub Actions, PyTest, Mockito, CI/CD, debugging automation
Tooling: Git/GitHub, JIRA, oscilloscopes, DMMs, Agile/Scrum, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLFQ scheduling** with timing metrics and visualization tooling, reducing debugging time by **20%** through deterministic test-driven simulation traces.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST pipeline** on AWS ECS/Fargate with automated CI/CD validation, improving uptime and production-grade reliability for language data processing workloads.

IoT Sensor Telemetry Pipeline | *Python, C++, MQTT, PostgreSQL*

- Developed a **low-latency telemetry ingestion system** for simulated edge sensors with anomaly checks and alerting, improving event traceability and fault isolation for embedded-style data streams.